# Mathematics HL – Analysis and Approaches

## Practice Paper 3

**Time allowed: 1 hour**

**Answer all the questions.**

All numerical answers must be given exactly or correct to three significant figures, unless otherwise stated in the question.

Answers should be supported by working and/or explanations.

Where an answer is incorrect, some marks may be awarded for a correct method, provided this is shown clearly.

**You need a graphic display calculator for this paper.**

| | Maximum marks | Marks awarded |
|---|---|---|
| Question 1 | 25 | |
| Question 2 | 35 | |
| **Total** | **60** | |

**Candidate name:** ______________________

**Candidate number:** ______________________

**Date:** ______________________

## Question 1 [25 marks]

Consider a sequence with first term $u_1$, defined by a recurrence relation of the form

$$u_{n+1} = au_n + b,$$

where $a$ and $b$ are constants.

For each of the given criteria in parts **a** and **b**:

**i.** state the type of sequence defined by the recurrence relation

**ii.** write down an expression for $u_n$ in terms of $u_1$ and $b$

**iii.** write down an expression for $S_n$ (the sum of the first $n$ terms) in terms of $u_1$ and $b$

**a** $a = 1$

**i.** Type: ____________________

**ii.** $u_n =$ ____________________

**iii.** $S_n =$ ____________________

**b** $b = 0, \quad a \neq 1, \quad a \neq 0$

**i.** Type: ____________________

**ii.** $u_n =$ ____________________

**iii.** $S_n =$ ____________________

**c** Consider the sequence defined by $u_1 = 3$ and $u_{n+1} = 2u_n - 1$.

**i.** Find $u_2, \; u_3, \; u_4$ and $u_5$.

| $n$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $u_n$ | 3 | | | | |

**ii.** Using your answers to part **i**, suggest an explicit expression for $u_n$ in terms of $n$.

$u_n =$ ____________________

**iii.** Prove your expression from part **ii** using mathematical induction.

*Let $P(n)$ be the statement:* $u_n =$ ______________________

**Base case** ($n = 1$):

**Inductive hypothesis:**
Assume $P(k)$ is true for some integer $k \geq 1$, i.e. $u_k =$ ______________________

**Inductive step** (show $P(k + 1)$ is true):

**Conclusion:**

**d** When $a \neq 1$, the recurrence relation $u_{n+1} = au_n + b$ can be rewritten in the form

$$u_{n+1} + c = a(u_n + c)$$

for some constant $c$.

**i.** Find the constant $c$ in terms of $a$ and $b$.

$c =$ ______________________________

A new sequence is defined by $w_n = u_n + c$.

**ii.** Write down the recurrence relation for the sequence $w_n$ and state what type of sequence $w_n$ is.

Recurrence for $w_n$: ______________________ Type: ____________________

**iii.** Hence, write down an explicit formula for $w_n$ in terms of $w_1$, $a$ and $n$.

$w_n =$ ______________________________________

**iv.** Hence, write down an explicit formula for $u_n$ in terms of $u_1$, $a$, $b$ and $n$.

$u_n =$ ______________________________________

**e** Verify that the expression you obtained for $u_n$ in part **d(iv)** for the sequence in part **c** is consistent with the formula you derived in part **c(ii)**.

**[Total: 25 marks]**

## Question 2 [35 marks]

In this question you will find the value of the indefinite integral

$$J = \int \frac{1}{\sqrt{4+x^2}}\,dx$$

using different methods.

Suppose that, first, you are asked to find the value of a definite integral. You can do this using technology.

**a** Write down the value of $\displaystyle\int_0^4 \frac{1}{\sqrt{4+x^2}}\,dx$.

$\displaystyle\int_0^4 \frac{1}{\sqrt{4+x^2}}\,dx =$ ____________________

Another integral, which looks similar to $J$, can be found using a trigonometric substitution.

**b** Write down $\displaystyle\int \frac{1}{\sqrt{1-x^2}}\,dx$.

$\displaystyle\int \frac{1}{\sqrt{1-x^2}}\,dx =$ ____________________________

**c** Show how the answer in **(b)** can be proved by applying the substitution $x = \sin\theta \left(-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}\right)$ to $\displaystyle\int \frac{1}{\sqrt{1-x^2}}\,dx$.

To find the integral $J$ by substitution, we are going to define two new functions:

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}.$$

**d** Show that $(\cosh x)^2 - (\sinh x)^2 = 1$.

**e** Find the derivative of **i** $\sinh x$ **ii** $\cosh x$, give your answers in terms of $\sinh x$ and $\cosh x$.

**i.** $\dfrac{d}{dx}(\sinh x) =$ ____________________________________

**ii.** $\dfrac{d}{dx}(\cosh x) =$ ____________________________________

The inverse function $\sinh^{-1} x$ is denoted by $\operatorname{arsinh} x$.

**f** By solving a quadratic in $e^x$, show and justify that

$$\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right).$$

**g** Find the indefinite integral $J$ by applying the substitution $x = 2\sinh u$.

$J =$ ____________________

**h** Hence, find the exact value of $\displaystyle\int_0^4 \frac{1}{\sqrt{4+x^2}}\,dx$ and verify that it agrees with the answer from part **(a)**.

Exact value: ____________ Decimal (3 s.f.): ____________ Agrees with (a)? ________

We will now attempt to see if we can find the integral $J$ just by using the usual trigonometrical functions.

**i** Apply the substitution $x = 2\tan\theta$, $\left(-\dfrac{\pi}{2} < \theta < \dfrac{\pi}{2}\right)$, to show that $J$ can be changed to $\displaystyle\int \sec\theta\, d\theta$.

**j** Multiply the $\sec\theta$ by 1, written as $\dfrac{\sec\theta+\tan\theta}{\sec\theta+\tan\theta}$, to obtain the answer to $\int \sec\theta\, d\theta$ in terms of $\theta$.

$\int \sec\theta\, d\theta =$ ______________________________

**k** Now express the answer found in part **j** in terms of $x$ and verify this is consistent with the expression you obtained for $J$ in part **g**.

You will now attempt to find the integral $J$ without using a substitution.

$$\text{Let } f(x) = x + \sqrt{4+x^2}.$$

**l** Find $f'(x)$.

$f'(x) =$ ______________________________________

**m** Multiply $\dfrac{1}{\sqrt{4+x^2}}$ by 1, written as $\dfrac{f'(x)}{f(x)}$, using your expression for $f'(x)$ from part **l**. Hence simplify the expression for $J$ into an expression involving only $f(x)$ and $f'(x)$. In this way, find the integral $J$ directly.

$J =$ ______________________________

**[Total: 35 marks]**

# MARK SCHEME & FULL WORKED ANSWERS

## QUESTION 1 [25 marks]

### Answer – 1a 3 marks

**i.** Arithmetic sequence
**ii.** $u_n = u_1 + (n-1)b$
**iii.** $S_n = \dfrac{n}{2}\big(2u_1 + (n-1)b\big)$

### Answer – 1b 3 marks

**i.** Geometric sequence (common ratio $a$)
**ii.** $u_n = u_1 \cdot a^{n-1}$
**iii.** $S_n = \dfrac{u_1(a^n - 1)}{a-1}$

### Answer – 1c 9 marks

**Part i:** Apply $u_{n+1} = 2u_n - 1$ repeatedly:

$$u_2 = 2(3) - 1 = 5, \quad u_3 = 2(5) - 1 = 9, \quad u_4 = 2(9) - 1 = 17, \quad u_5 = 2(17) - 1 = 33$$

**Part ii:** The terms 3, 5, 9, 17, 33 are each one more than a power of 2:

$$3 = 2^1 + 1, \quad 5 = 2^2 + 1, \quad 9 = 2^3 + 1, \quad 17 = 2^4 + 1, \quad 33 = 2^5 + 1$$

$$\boxed{u_n = 2^n + 1}$$

**Part iii — Proof by Mathematical Induction:**

**Base case ($n = 1$):**

$u_1 = 2^1 + 1 = 3$ ✓ The base case holds.

**Inductive hypothesis:**

Assume $P(k)$ is true for some integer $k \geq 1$: $u_k = 2^k + 1$.

**Inductive step (prove $P(k+1)$, i.e. $u_{k+1} = 2^{k+1} + 1$):**

Using the recurrence relation and then the hypothesis:

$$\begin{aligned} u_{k+1} &= 2u_k - 1 \\ &= 2(2^k + 1) - 1 \qquad \text{[by inductive hypothesis]} \\ &= 2^{k+1} + 2 - 1 \\ &= 2^{k+1} + 1 \quad \checkmark \end{aligned}$$

This is exactly $P(k+1)$.

**Conclusion:**

Since $P(1)$ is true, and $P(k) \Rightarrow P(k+1)$ for all integers $k \geq 1$, by the principle of mathematical induction $P(n)$ is true for all integers $n \geq 1$.

**Answer – 1e** **2 marks**

For the sequence in part **c**: $a = 2$, $b = -1$, $u_1 = 3$.
Substitute into the formula from d(iv):

$$u_n = \left(3 + \frac{-1}{2-1}\right) 2^{n-1} - \frac{-1}{2-1} = (3-1) \cdot 2^{n-1} + 1 = 2 \cdot 2^{n-1} + 1 = 2^n + 1 \quad \checkmark$$

This is consistent with the formula derived in part c(ii).

## QUESTION 2  [35 marks]

### Answer – 2a  2 marks

Using a GDC: $\displaystyle\int_0^4 \frac{1}{\sqrt{4+x^2}}\,dx \approx 1.44 \quad \text{(3 s.f.)}$

### Answer – 2b  1 mark

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin x + C$$

### Answer – 2c  4 marks

Let $x = \sin\theta$. Then $\dfrac{dx}{d\theta} = \cos\theta$, so $dx = \cos\theta\,d\theta$.
Using the Pythagorean identity:

$$1 - x^2 = 1 - \sin^2\theta = \cos^2\theta \;\Rightarrow\; \sqrt{1-x^2} = \cos\theta \quad \left(\text{positive since } -\tfrac{\pi}{2} \le \theta \le \tfrac{\pi}{2}\right)$$

Substituting:

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = \int \frac{1}{\cos\theta}\cdot\cos\theta\,d\theta = \int 1\,d\theta = \theta + C = \arcsin x + C \quad \checkmark$$

### Answer – 2d  2 marks

$$\begin{aligned}\cosh^2 x - \sinh^2 x &= \left(\frac{e^x+e^{-x}}{2}\right)^2 - \left(\frac{e^x-e^{-x}}{2}\right)^2\\ &= \frac{e^{2x}+2+e^{-2x}}{4} - \frac{e^{2x}-2+e^{-2x}}{4}\\ &= \frac{(e^{2x}+2+e^{-2x})-(e^{2x}-2+e^{-2x})}{4}\\ &= \frac{4}{4} = 1 \quad \checkmark\end{aligned}$$

### Answer – 2e  4 marks

**Part i:**

$$\frac{d}{dx}(\sinh x) = \frac{d}{dx}\left(\frac{e^x-e^{-x}}{2}\right) = \frac{e^x+e^{-x}}{2} = \cosh x$$

**Part ii:**

$$\frac{d}{dx}(\cosh x) = \frac{d}{dx}\left(\frac{e^x + e^{-x}}{2}\right) = \frac{e^x - e^{-x}}{2} = \sinh x$$

## Answer – 2f 5 marks

Let $y = \operatorname{arsinh} x$. By definition, $x = \sinh y = \dfrac{e^y - e^{-y}}{2}$.
**Step 1:** Multiply both sides by $2e^y$:

$$2x = e^y - e^{-y} \;\Rightarrow\; 2xe^y = e^{2y} - 1 \;\Rightarrow\; e^{2y} - 2xe^y - 1 = 0$$

**Step 2:** This is a quadratic in $e^y$. Let $t = e^y$: $t^2 - 2xt - 1 = 0$.
**Step 3:** Quadratic formula:

$$t = \frac{2x \pm \sqrt{4x^2+4}}{2} = x \pm \sqrt{x^2+1}$$

**Step 4 (justify):** Since $t = e^y > 0$ for all $y$, and $\sqrt{x^2+1} > |x|$ implies $x - \sqrt{x^2+1} < 0$, we must take the positive root:

$$e^y = x + \sqrt{x^2+1}$$

**Step 5:** Taking natural logs:

$$\boxed{y = \operatorname{arsinh} x = \ln\left(x + \sqrt{x^2+1}\right)} \quad \checkmark$$

## Answer – 2g 4 marks

Let $x = 2\sinh u$. Then $\dfrac{dx}{du} = 2\cosh u$, so $dx = 2\cosh u\, du$.
Using the identity from part **d**:

$$4 + x^2 = 4 + 4\sinh^2 u = 4\cosh^2 u \;\Rightarrow\; \sqrt{4+x^2} = 2\cosh u$$

Substituting:

$$J = \int \frac{1}{2\cosh u} \cdot 2\cosh u\, du = \int 1\, du = u + C$$

Converting back using $x = 2\sinh u \Rightarrow u = \operatorname{arsinh}(x/2) = \ln\left(\dfrac{x}{2} + \sqrt{\dfrac{x^2}{4} + 1}\right)$:

$$\boxed{J = \ln\left(x + \sqrt{x^2+4}\right) + C}$$

(The constant $\ln 2$ is absorbed into $C$.)

**Answer – 2h** **2 marks**

$$\int_0^4 \frac{dx}{\sqrt{4+x^2}} = \left[\operatorname{arsinh}\tfrac{x}{2}\right]_0^4 = \operatorname{arsinh}(2) - \operatorname{arsinh}(0) = \ln\!\left(2+\sqrt{5}\right) - 0 = \ln\!\left(2+\sqrt{5}\right)$$

$$\ln(2+\sqrt{5}) \approx 1.44 \quad \checkmark \quad \text{(consistent with part a)}$$

**Answer – 2i** **2 marks**

Let $x = 2\tan\theta$. Then $dx = 2\sec^2\theta\,d\theta$.
Using the identity $1+\tan^2\theta = \sec^2\theta$:

$$4+x^2 = 4+4\tan^2\theta = 4\sec^2\theta \;\Rightarrow\; \sqrt{4+x^2} = 2\sec\theta$$

Substituting:

$$J = \int \frac{1}{2\sec\theta}\cdot 2\sec^2\theta\,d\theta = \int \sec\theta\,d\theta \quad \checkmark$$

**Answer – 2j** **2 marks**

$$\int \sec\theta\,d\theta = \int \sec\theta\cdot\frac{\sec\theta+\tan\theta}{\sec\theta+\tan\theta}\,d\theta = \int \frac{\sec^2\theta+\sec\theta\tan\theta}{\sec\theta+\tan\theta}\,d\theta$$

The numerator is exactly the derivative of the denominator, so:

$$\int \frac{\sec^2\theta+\sec\theta\tan\theta}{\sec\theta+\tan\theta}\,d\theta = \ln|\sec\theta+\tan\theta| + C$$

**Answer – 2k** **2 marks**

From $x = 2\tan\theta$: $\quad \tan\theta = \dfrac{x}{2} \quad$ and $\quad \sec\theta = \dfrac{\sqrt{4+x^2}}{2}$
Substituting:

$$J = \ln\left|\frac{\sqrt{4+x^2}}{2} + \frac{x}{2}\right| + C = \ln\left(\frac{x+\sqrt{4+x^2}}{2}\right) + C = \ln\!\left(x+\sqrt{4+x^2}\right) + C'$$

(absorbing $-\ln 2$ into $C'$) $\checkmark$ Consistent with part **g**.

**Answer – 2l** **2 marks**

$$f(x) = x+\sqrt{4+x^2} = (x) + (4+x^2)^{1/2}$$

$$f'(x) = 1 + \frac{1}{2}(4+x^2)^{-1/2}\cdot 2x = 1 + \frac{x}{\sqrt{4+x^2}} = \frac{\sqrt{4+x^2}+x}{\sqrt{4+x^2}}$$

**Answer – 2m** **3 marks**

From part **l**:

$$\frac{f'(x)}{f(x)} = \frac{\dfrac{\sqrt{4+x^2}+x}{\sqrt{4+x^2}}}{x+\sqrt{4+x^2}} = \frac{\sqrt{4+x^2}+x}{\sqrt{4+x^2}} \cdot \frac{1}{x+\sqrt{4+x^2}} = \frac{1}{\sqrt{4+x^2}}$$

Therefore:

$$J = \int \frac{1}{\sqrt{4+x^2}}\,dx = \int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C$$

$$\boxed{J = \ln\left(x+\sqrt{4+x^2}\right) + C} \quad \checkmark$$

## FINAL ANSWER SUMMARY

| Part | Key result | Marks |
|---|---|---|
| 1a | Arithmetic; $u_n = u_1 + (n-1)b$; $S_n = \frac{n}{2}(2u_1 + (n-1)b)$ | 3 |
| 1b | Geometric; $u_n = u_1 a^{n-1}$; $S_n = \frac{u_1(a^n-1)}{a-1}$ | 3 |
| 1c | $u_n = 2^n + 1$ (proved by induction) | 9 |
| 1d | $c = \frac{b}{a-1}$; $u_n = \left(u_1 + \frac{b}{a-1}\right) a^{n-1} - \frac{b}{a-1}$ | 8 |
| 1e | Substituting $a = 2, b = -1, u_1 = 3$ gives $u_n = 2^n + 1$ ✓ | 2 |
| **Question 1 Total** | | **25** |
| 2a | $\approx 1.44$ (GDC) | 2 |
| 2b | $\arcsin x + C$ | 1 |
| 2c | Substitution $x = \sin\theta$ gives $\int d\theta = \arcsin x + C$ | 4 |
| 2d | $\cosh^2 x - \sinh^2 x = 1$ (shown by expanding definitions) | 2 |
| 2e | $\frac{d}{dx}(\sinh x) = \cosh x$; $\frac{d}{dx}(\cosh x) = \sinh x$ | 4 |
| 2f | $\operatorname{arsinh} x = \ln(x + \sqrt{x^2+1})$ (quadratic in $e^y$) | 5 |
| 2g | $J = \ln(x + \sqrt{x^2+4}) + C$ (via $x = 2\sinh u$) | 4 |
| 2h | $\ln(2+\sqrt{5}) \approx 1.44$ ✓ | 2 |
| 2i | $x = 2\tan\theta$ gives $\int \sec\theta \, d\theta$ | 2 |
| 2j | $\int \sec\theta \, d\theta = \ln\lvert\sec\theta + \tan\theta\rvert + C$ | 2 |
| 2k | Converts to $\ln(x + \sqrt{4+x^2}) + C$ ✓ | 2 |
| 2l | $f'(x) = \frac{x+\sqrt{4+x^2}}{\sqrt{4+x^2}}$ | 2 |
| 2m | $J = \ln(x + \sqrt{4+x^2}) + C$ (via $f'/f$) | 3 |
| **Question 2 Total** | | **35** |
| **PAPER TOTAL** | | **60** |